

CrowdSense

Real-time crowd formation & risk assess

Abstract

CrowdSense is a real-time framework for detecting emergent crowd gatherings in CCTV-like video and generating near-term safety risk assessments that support operational decision-making in smart city control settings. The system uses density-map-based crowd counting to estimate human presence under occlusion and scale variation, avoiding brittle person detection and tracking in congested scenes. Each frame is converted into a crowd density map (via DM-Count) and summarized into a compact set of “gathering signals” capturing crowd size and spatial structure (e.g., total count, hotspot concentration, spread, and coarse zone occupancy).

For risk assessment, CrowdSense introduces a two-phase LLM reasoning layer. Phase 1 performs single-frame analysis to produce an operator-style status label (Sparse, Moderate, Crowded, Heavily Crowded) with a short justification derived strictly from the extracted signals. When the status exceeds Sparse, Phase 2 is triggered: the model receives a short time-ordered sequence of recent frame summaries (“latest events”) and outputs a risk assessment report that prioritizes escalation patterns (e.g., surge vs. compression/locking) and recommends immediate actions. By combining density-driven measurement with structured, interpretable LLM reporting over both instantaneous and short-term temporal context, CrowdSense aims to provide early-warning cues before crowd build-up evolves into hazardous conditions

Introduction

Motivation and Problem Statement

Large public gatherings in smart cities (e.g., concerts, demonstrations, sporting events) can rapidly shift from organized activity to high-risk situations, including overcrowding and violent escalation. City control rooms often rely on manual monitoring across many camera feeds - an approach that is slow, subjective, and difficult to scale - creating a need for operational early-warning capabilities that detect emerging gatherings and support timely, proportionate intervention when risk indicators begin to rise.

CrowdSense is motivated by two operational needs: (1) detecting when a gathering is forming so relevant authorities can assess legitimacy via policy and external channels, and (2) assessing near-term crowd risk (e.g., over-density, stalled flow, turbulent motion) to guide response. Technically, explicit person detection and tracking are fragile in surveillance conditions due to occlusions, perspective effects, and dense congestion. Density-based crowd

analysis offers a practical alternative by estimating both crowd size and spatial concentration from continuous density maps, even when individuals cannot be reliably separated.

Research Question

Can emergent visual signatures in CCTV surveillance be used to detect the formation of spontaneous gatherings and assess their escalation into dangerous situations?

CrowdSense Overview and Contributions

CrowdSense is a real-time framework designed to support operational monitoring by converting raw CCTV video into interpretable signals about crowd formation and short-horizon escalation risk. Instead of relying on explicit detection and tracking of individuals, CrowdSense adopts a density-based representation of human presence: each frame is mapped to a continuous crowd density estimate, enabling robust measurement under occlusion and congestion. The density output is then summarized into compact “gathering signals” that capture both volume (how many people) and structure (how concentrated, how spread, and where the crowd is localized). This standardized representation enables consistent monitoring across camera views and provides a structured input for downstream reasoning.

Building on these signals, CrowdSense performs risk reasoning in two stages. First, a single-frame analysis produces an operator-style status label (e.g., Sparse to Heavily Crowded) with a short justification grounded in the extracted indicators. When the status indicates a meaningful gathering, the system then analyzes a short sequence of recent frame summaries to identify escalation patterns such as rapid influx, increasing concentration/compaction, or constrained crowd geometry. It outputs a structured risk assessment report that highlights the primary drivers behind the assessed risk level and recommends immediate operator actions, prioritizing interpretability and operational usefulness.

In summary, the main contributions of this work are: (1) a density-driven pipeline for real-time gathering detection that avoids fragile per-person tracking in congested scenes; (2) a standardized, interpretable feature representation derived from density maps for both per-frame and temporal reasoning; and (3) a two-phase LLM-based decision layer that transforms these signals into operator-facing status and risk reports with explicit drivers and actionable guidance.

Literature Review and Existing Approaches

Detecting human gatherings in public spaces is central to urban safety, public health, and event monitoring. In real surveillance environments, however, explicit person detection and multi-object tracking often fail under the very conditions that matter most: partial occlusions, scale variation, and heavy congestion. As a result, a major line of work has shifted toward **density-based crowd analysis**, where people are modeled as a continuous spatial density distribution rather than a set of independently detected objects [2]. Density maps make it possible to estimate **how many** people are present and **where** they are concentrated even

when individual instances cannot be reliably separated, forming the basis of modern crowd counting and downstream scene reasoning [1,2].

Density-Based Crowd Counting

Most modern crowd counting methods rely on **density map regression**, where convolutional networks predict a continuous density field whose integral yields the crowd count. CSRNet introduced a strong baseline for highly congested scenes by showing that dilated convolutions can produce accurate density estimates under complex perspective and occlusion [2]. DM-Count advanced this paradigm by framing learning as **distribution matching** between predicted density and point-level annotations, resulting in sharper localization and strong counting performance across standard benchmarks [1]. In the context of real-time monitoring, these methods provide a robust measurement layer that can operate where detection and tracking degrade.

Limitation for safety monitoring: crowd counting models primarily output *counts* (and sometimes density fields) but do not directly answer higher-level operational questions such as whether a **gathering event** is forming or whether the crowd's state is **escalating toward risk**. Bridging this gap requires additional structure on top of density estimation.

From Density Maps to Gathering Detection

Density maps also implicitly support localization and hotspot reasoning: peaks and local maxima can act as proxies for head locations or clustered subregions [1,2]. This enables downstream tasks such as estimating **per-region occupancy**, identifying crowded subareas, and tracking where density concentrates over time - capabilities that are useful for defining “gatherings” without requiring explicit per-person tracks.

Key gap: while prior density-based work provides the raw signals (counts and spatial density), it often stops short of offering an **operational definition of gathering formation** that is immediately usable in control-room settings (e.g., zone-level summaries and standardized indicators that can be monitored consistently across cameras).

Crowd Risk, Congestion, and Safety Indicators

A distinct body of literature emphasizes that the main hazard in crowd disasters is not merely the presence of a crowd, but the transition into dangerous dynamics as density rises and movement becomes restricted. Empirical and modeling studies show that extreme density in confined geometries can produce stop-and-go waves and turbulent motion that precede crush incidents [3]. Building on this principle, Bek and Monari proposed a video-based **crowd congestion level** that explicitly couples local density with reduced motion to quantify risk in surveillance footage [4]. These works motivate risk indicators that jointly reflect **crowd volume**, **spatial compaction**, and **constraints on flow** - signals that are more aligned with safety outcomes than count alone.

Limitation: many safety indicators require either explicit motion estimation or scene-specific calibration, and they are not always presented in a form that supports rapid operator interpretation across multiple live feeds.

Learning-Based Anomaly and Risk Prediction from Crowd Features

More recent work explores learning-based models that operate on **crowd-level features** to detect or predict hazardous conditions. The SIMCD framework, for example, uses simulated crowd data with severity and abnormality labels to train both unsupervised anomaly detectors and supervised temporal models (e.g., sequence models) for early warning of dangerous configurations [5]. This line of work supports the broader claim that short histories of crowd features - density, speed, and directional consistency - can be sufficient to forecast escalation.

Practical challenges: in real surveillance deployments, labeled incident data is scarce, environments vary widely, and purely model-based risk scores can be difficult to trust without **interpretable drivers**. This motivates approaches that combine (i) robust feature extraction, (ii) temporal reasoning over short windows, and (iii) outputs that explain *why* a risk assessment was produced.

Summary of the Gap and Positioning of CrowdSense

Across these strands, prior work has established how to (1) obtain reliable density maps from single frames [1,2], (2) link high density and constrained movement to critical crowd dynamics [3,4], and (3) use crowd-level features for anomaly detection and forecasting [5]. However, an operational monitoring system still needs a cohesive bridge between these components: **standardized, camera-agnostic gathering signals** derived from density maps, plus a **temporal escalation layer** that produces interpretable, operator-facing outputs.

CrowdSense is positioned to connect these pieces in a single workflow: a density-map backbone (DM-Count) provides robust per-frame spatial measurements [1], which can be summarized into geometry-aware indicators inspired by safety literature [3,4] and then used for temporal escalation reasoning consistent with feature-based forecasting approaches [5]. The emphasis is not only on predicting risk, but on making the assessment traceable to concrete measurable factors (e.g., rising density, increasing hotspot dominance, compaction, and stagnation), enabling practical early-warning support in smart city surveillance settings

Methodology

System overview

CrowdSense is a real-time pipeline that transforms surveillance video into structured crowd signals and operator-facing alerts. The system follows a modular **Sense–Detect–Predict** flow: video is sampled into frames, each frame is converted into a crowd density representation, density is summarized into compact features, and these features are consumed by an LLM-based reasoning layer to produce status and risk reports for a dashboard interface.

This separation enables (i) robust visual measurement under occlusion and congestion, (ii) a standardized, camera-agnostic feature interface, and (iii) interpretable decision support that can incorporate both instantaneous and short-term temporal evidence.

Density estimation backbone

CrowdSense uses a density-map crowd counting model as its perception backbone. In this paradigm, the model predicts a continuous density field over the image rather than explicitly detecting and tracking each person, which is advantageous in crowded scenes where individuals are difficult to separate. The implementation is based on DM-Count with a convolutional feature extractor (VGG-style backbone) and a density prediction head; pretrained weights can be used and optionally adapted to the deployment domain via fine-tuning on standard crowd-counting datasets.

Density-to-feature representation

For each frame, the predicted density map is converted into a standardized set of “gathering signals” serialized as a compact JSON record. The feature schema is designed to capture both **crowd size** and **crowd structure**, and to be usable for single-frame interpretation as well as temporal trend analysis. The main feature families are:

- **Volume (how many):** a count estimate derived by integrating the density map, used for high-level status reporting and alerting.
- **Concentration (how packed):** distributional indicators such as entropy-like uniformity measures and “top-percent mass” summaries that quantify how much of the crowd is concentrated in the densest regions; optional absolute counts for the densest areas support hotspot-specific alerts.
- **Spatial geometry (where and what shape):** center-of-mass location, spread/radius, and anisotropy/elongation measures that can indicate corridor-, queue-, or lane-like configurations.
- **Zone/grid summaries (where within the scene):** a coarse grid partition over the frame, per-cell mass, and “top-k cell mass” summaries to capture localized crowding and uneven distribution at an operationally meaningful resolution.

Each per-frame record can be logged into a lightweight time series (“latest events”), enabling risk reasoning over short temporal windows without storing full video.

LLM Phase 1: single-frame status detection

Phase 1 provides a per-frame operational summary intended for fast operator consumption. The LLM receives a single-frame JSON feature record and returns a fixed-format output containing: (i) a discrete crowd **status label** (e.g., Sparse/Moderate/Crowded/Heavily Crowded), (ii) a short justification grounded strictly in the provided metrics, and (iii) an initial recommended action.

The labeling logic is **threshold-based**, but the thresholds are treated as configurable policy parameters (e.g., per camera or per environment). In addition to global volume, the prompt enforces explicit mention of localized hotspots when densest-area indicators exceed configured significance levels, so that localized compression is not hidden by an “average” crowd count.

LLM Phase 2: temporal risk assessment over recent events

Phase 2 is triggered only when Phase 1 indicates that the scene contains a meaningful gathering (i.e., above a minimal significance threshold). The LLM receives a short, time-ordered sequence of recent feature records (“latest events”) and produces a structured **risk assessment report** with immediate operator actions.

Temporal risk reasoning prioritizes escalation mechanisms that are consistent with crowd safety theory and operational monitoring needs. The prompt is designed around a hierarchical trigger logic, where certain high-risk patterns override simple volume-based alerts. The main trigger families include:

- **Locking / compression patterns:** detected as sustained compaction signals over consecutive frames (e.g., increasing concentration and/or decreasing uniformity together with strong directional/elongated structure), indicating a transition toward constrained flow and potential crush risk.
- **Surge patterns:** identified by rapid growth in crowd volume over a short period, even if density is not yet extreme.
- **Volume patterns:** high overall occupancy and/or persistent high-density hotspots, used as a baseline severity signal when more specific escalation signatures are absent.

The output report includes: (i) a relative status trend (stable/escalating/de-escalating), (ii) the primary trigger category, (iii) concise driver explanations grounded in the observed feature trends, (iv) recommended immediate actions, and (v) a compact summary table of recent key indicators for transparency.

Deployment and operational considerations

The methodology is designed for real-time deployment but acknowledges practical constraints: runtime depends on frame resolution and model throughput; crowd/risk thresholds may require calibration to local camera geometry and operational policy; and LLM latency/cost can be significant. To manage compute, CrowdSense uses a gated strategy where temporal risk reporting is only invoked when a non-trivial crowd is detected, and the temporal window is restricted to the most recent events needed for escalation reasoning.

Experiments and Evaluation

Experimental Setup

<TBD>

Datasets

<TBD>

Baselines and Comparisons

<TBD>

Metrics

<TBD>

Results

<TBD>

Ablation / Sensitivity

<TBD>

References

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